

ASSIGNMENT 3

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Problem 1:

Write a Java program to check whether a given number is Prime or Not.

- A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Expected Input & Output

Input: 7

Output: 7 is a Prime number

Input: 12

Output: 12 is Not a Prime number

Constraints:

Input must be an integer greater than 1.

Program should handle large numbers efficiently.

Hints:

Hint 1: Use a loop to check divisibility from 2 to $n/2$.

Hint 2: If divisible, it is not a prime. Else, it is prime.

Program:

```
package Assignment;

import java.util.Scanner;

public class PrimeNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        for (int j = 1; j <= 2; j++) {

            System.out.print("Enter integer " + j + ": ");

            int n = sc.nextInt();

            boolean prime = true;

            if (n <= 1) {

                prime = false;

            } else {
```

```

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
    }
    if (prime) {
        System.out.println(n + " is a Prime number");
    } else {
        System.out.println(n + " is Not a Prime number");
    }
}
sc.close();
}
}

```

Output:

Enter integer 1: 7

7 is a Prime number

Enter integer 2: 12

12 is Not a Prime number

- This program checks whether the entered numbers are Prime numbers or not.
- Scanner is used to take user input from the keyboard.
- A for loop is used to accept and check two numbers.
- Stored the input number in variable n.
- A boolean variable prime is initially set to true.
- If the number is less than or equal to 1, it is marked as not prime.

- Another loop checks divisibility from 2 to $n/2$.
- If the number is divisible by any value, prime becomes false and the loop breaks.
- Finally, the program prints whether the number is prime or not.
- At the end, the Scanner is closed using `sc.close()`.

Problem 2:

Write a Java program to find the largest and smallest element in an array.

Expected Input & Output

Input: {12, 45, 2, 67, 34}

Output: Largest = 67, Smallest = 2

Constraints:

Array should have at least one element.

Handle both positive and negative numbers.

Hints:

Hint 1: Assume first element as both largest and smallest.

Hint 2: Traverse the array and update values accordingly.

Program:

```
package Assignment;

public class LargestSmallest {

    public static void main(String[] args) {

        int[] arr = {12, 45, 2, 67, 34};
        int largest = arr[0];
        int smallest = arr[0];

        for(int i = 1; i < arr.length; i++) {
            if(arr[i] > largest) {
                largest = arr[i];
            }
            if(arr[i] < smallest) {
                smallest = arr[i];
            }
        }
    }
}
```

```
        System.out.println("Largest = " + largest);  
        System.out.println("Smallest = " + smallest);  
    }  
}
```

Output:

Largest = 67

Smallest = 2

- This program finds the Largest and Smallest number in an array.
- An integer array is created with some values.
- Both largest and smallest are initially assigned as the first array element.
- A for loop is used to check each element in the array one by one.
- If the current element is greater than largest, it updates the largest value.
- If the current element is smaller than smallest, it updates the smallest value.
- Likewise, each value is compared and updated accordingly.
- Finally, the program prints the largest and smallest numbers.

Problem 3:

Write a Java program to count the number of vowels and consonants in a string.

Expected Input & Output

Input: Hello World

Output: Vowels = 3, Consonants = 7

Constraints:

String may contain spaces or punctuation (ignore them).

Treat both uppercase and lowercase letters the same.

Hints:

Hint 1: Convert string to lowercase for easy comparison.

Hint 2: Use charAt() to traverse characters.

Hint 3: Check if the character belongs to vowels (a, e, i, o, u).

Program:

```
package Assignment;

import java.util.Scanner;

public class VowelsConsonants {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Input: ");

        String str = sc.nextLine().toLowerCase();

        int vowels = 0, consonants = 0;

        for(int i=0; i< str.length(); i++) {

            char ch = str.charAt(i);

            if(ch>='a' && ch<='z') {

                if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u') {
```

```

        vowels++;
    } else {
        consonants++;
    }
}

System.out.print("Output: Vowels = " + vowels + ", Consonants = " +
consonants);

sc.close();
}
}

```

Output:

Input: Hello World

Output: Vowels = 3, Consonants = 7

- This program counts the number of Vowels and Consonants in a given string input.
- Scanner is used to take user input from the keyboard.
- The input string is converted to lowercase using toLowerCase() to make vowel checking easier and case-insensitive.
- Two variables, vowels and consonants, are initialised to store the counts.
- A for loop is used to traverse each character in the string one by one.
- charAt(i) is used to extract each character from the string.
- The if condition `ch >= 'a' && ch <= 'z'` checks whether the character is an alphabet.
- The next if condition checks whether the character is a vowel (a, e, i, o, u).
- If it's a vowel, the vowel counter is increased; otherwise, the consonant counter is increased.
- Finally, the program prints the total number of vowels and consonants.